

LINCOLN 11 SW, NE, USA

WMO: 744410

Lat: 40.6954N

Lon: 96.8541W

Elev: 1372

StdP: 13.98

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 94996

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -1.9	(c) 2.9	(d) -10.3	(e) 3.3	(f) 0.0	(g) -5.7	(h) 4.3	(i) 3.7	(j) 18.7	(k) 31.4	(l) 17.4	(m) 30.4	(n) 5.1	(o) N/A	(p) 0.471

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 20.1	(c) 94.7	(d) 73.9	(e) 91.5	(f) 73.8	(g) 88.7	(h) 73.1	(i) 78.1	(j) 88.8	(k) 76.5	(l) 86.9	(m) 74.9	(n) 85.0	(o) 8.1	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.1	138.5	85.3	73.2	129.9	83.2	71.6	122.8	81.6	42.7	88.7	40.8	86.8	39.2	84.9	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.9	14.9	13.3		-9.5	99.3	2.9	3.0	-11.6	101.4	-13.3	103.2	-14.9	104.8	-17.1	107.0	
				DB	-10.0	80.4	3.4	1.6	-12.4	81.5	-14.4	82.4	-16.4	83.3	-18.8	84.4
				WB	-10.0	80.4	3.4	1.6	-12.4	81.5	-14.4	82.4	-16.4	83.3	-18.8	84.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.0	25.5	28.2	41.9	52.0	62.0	72.3	76.4	74.0	67.0	53.5	40.7	28.8
	DBStd	20.39	12.38	12.40	12.15	10.18	8.70	6.52	6.11	5.87	8.27	9.83	10.59	10.98
	HDD50	2835	761	614	310	97	10	0	0	0	1	75	310	657
	HDD65	5859	1226	1030	719	403	162	13	3	5	74	375	728	1121
	CDD50	3561	0	4	59	156	383	668	818	744	512	184	32	1
	CDD65	1108	0	0	2	12	70	231	356	284	134	19	0	0
	CDH74	11126	0	0	42	173	721	2283	3702	2695	1301	200	9	0
	CDH80	4364	0	0	7	42	214	854	1629	1093	473	51	1	0
Wind	WSAvg	6.3	6.7	7.0	7.5	7.9	6.5	5.6	4.9	4.7	5.5	6.1	6.6	6.4
Precipitation	PrecAvg	28.7	0.7	0.8	1.7	2.6	4.7	4.5	4.2	3.1	2.8	2.1	1.5	0.8
	PrecMax	42.3	1.3	2.4	3.7	4.4	10.6	8.6	12.1	6.7	4.9	5.2	3.1	2.6
	PrecMin	21.2	0.0	0.1	0.0	0.6	1.2	0.2	1.0	1.0	0.4	0.1	0.1	0.0
	PrecStd	5.5	0.4	0.6	1.0	0.9	2.7	1.9	2.6	1.5	1.3	1.4	0.9	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.2	68.0	78.8	85.4	90.8	95.2	100.1	97.8	93.9	86.3	74.2	59.7
		MCWB	45.4	49.1	59.1	65.1	67.8	74.6	72.4	72.5	71.2	64.8	56.1	47.8
	2%	DB	52.3	58.2	72.9	79.0	85.3	91.4	95.4	93.3	89.4	79.0	67.3	53.7
		MCWB	41.6	47.7	54.9	60.7	67.3	74.2	75.0	73.0	71.4	61.2	53.8	43.9
	5%	DB	46.7	51.3	67.1	73.9	81.8	88.2	92.1	89.5	86.0	74.9	61.8	49.0
		MCWB	38.8	42.1	53.1	57.6	65.8	73.2	75.8	73.6	70.1	59.1	50.4	41.5
	10%	DB	41.9	45.4	61.2	69.2	77.8	85.1	88.6	86.2	81.6	70.0	56.9	44.1
		MCWB	36.4	39.2	50.3	55.1	63.8	71.7	75.4	72.3	67.8	56.6	47.5	38.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.6	54.5	63.5	65.1	72.9	78.9	80.8	79.6	75.2	68.3	59.7	53.2
		MCDB	56.7	61.7	76.7	78.2	83.6	89.8	90.8	90.2	88.6	77.4	66.9	57.3
	2%	WB	42.4	48.1	59.3	62.2	70.0	76.6	78.5	77.2	73.4	64.2	55.3	46.1
		MCDB	50.3	59.7	68.4	73.9	80.7	87.4	89.1	87.6	85.5	73.9	64.2	52.5
	5%	WB	39.4	42.7	54.9	59.6	68.4	75.0	77.1	75.2	72.0	61.4	51.7	42.0
		MCDB	45.7	50.9	65.8	69.4	79.8	85.3	87.6	85.0	82.7	70.9	60.1	49.0
	10%	WB	36.2	39.1	51.1	56.9	66.2	73.3	75.4	73.4	70.3	58.4	48.3	38.4
		MCDB	41.4	45.2	61.7	66.2	76.5	83.2	85.1	82.6	79.3	67.0	56.5	44.1
Mean Daily Temperature Range	5% DB	MDBR	18.2	19.2	21.9	22.2	20.9	19.9	20.1	21.0	22.5	22.1	20.8	17.7
		MCDBR	25.7	28.0	30.1	29.1	25.7	22.4	23.4	24.5	24.8	27.5	27.4	25.0
	5% WB	MCWBR	17.0	17.8	16.9	15.2	11.2	10.8	9.5	10.2	10.6	14.4	15.4	16.8
		MCWBR	23.7	26.8	27.2	24.2	22.1	20.6	19.4	20.7	20.9	23.4	24.6	23.6
Clear-Sky Solar Irradiance	taub		0.281	0.298	0.322	0.374	0.402	0.412	0.419	0.418	0.379	0.331	0.293	0.278
	taud		2.475	2.445	2.417	2.294	2.286	2.314	2.293	2.300	2.359	2.481	2.513	2.486
	Ebn at Noon		282	292	296	285	278	274	271	268	272	276	275	274
	Edh at Noon		24	29	33	40	42	41	41	40	35	28	23	22
All-Sky Solar Radiation	RadAvg		679	959	1262	1595	1872	2103	2089	1825	1492	1068	756	573
	RadStd		58	60	105	103	135	138	128	96	98	115	53	55

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page